

Turing Decay: A Three-Factor Model of Human Detection of AI-Generated Content, with Preregistered Predictions for Audio and Video

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Abstract

How fast is people’s ability to tell AI-generated content from real content disappearing, and what drives the decline? We separate two clocks that earlier work has blended: when the *generator* was released, and when the *study* was run. In a pilot, we re-analyze 50 face experiments from a 2026 meta-analysis and 17 text results from 9 studies. For faces, accuracy fell about 5 percentage points for each year of newer generator release within the GAN lineage, while five years of studies against the same generator showed no improvement. The first text-to-image generators reset detectability upward before it decayed again, so the decline looks like a sawtooth rather than a single curve. For text, untrained readers were at chance by GPT-3 (2020); holding release date fixed, accuracy fell 6.5 points per tenfold increase in model size, so capability, not calendar time, drives the decline. In one small study, frequent users of language models detected AI text about 95% of the time, and a humanlike persona prompt cut Turing-test detection from 79% to 27%.

From these results we propose Turing Decay, a three-factor model: generator capability and the effort spent deploying a generator push human accuracy toward and below chance, and hands-on expertise may push back. We preregistered six predictions for audio and video and tested them in a meta-analysis of 125 participant groups from 58 studies, screened from 7,609 records. That test is consistent with the model but does not confirm it. People still detect AI-generated audio and video above chance: by model estimates, about 12 percentage points for the newest audio generators and 16 for the newest video generators, down from about 17 for generators released in 2020. For audio, sensitivity (d' , a measure of discrimination that is not affected by a tendency to answer “real” or “fake”; a secondary measure, not the registered primary test) falls 0.23 per year of generator release; the accuracy slope (-1.3 points per year) points the same way but is not significant and shrinks when influential groups are dropped. Video shows no decline overall; an exploratory fit within text-to-video generators suggests the margin above chance halves in about 1.3 years. No directional prediction survives the registered correction for multiple tests, and deployment effort, expertise and family resets are still too rarely studied to test. The claim the evidence supports is narrower than the model: within a generator lineage, detection falls as generators improve and does not recover with calendar time, and hands-on expertise may counteract it.

Highlights

- Detection tracks the generator’s release date, not the year a study was run.
- Newer generators are harder to spot; exposure gave no detectable gain.
- People still beat chance on AI audio and video, by about 12–16 points today.

- Audio sensitivity (d' , a secondary measure) falls 0.23 per year of newer generator.
- Audio and video results are consistent with the model but not confirmatory.

1 Introduction

Moore’s law described a regular trend in one quantity, transistor density, and that regularity made it useful for planning. We ask whether a similarly regular trend describes a quantity that matters as much for the information ecosystem: how well people can tell AI-generated content from real content.

Many individual studies now report human detection near chance, for faces, voices, video and text. Two meta-analyses have pooled these results and asked whether humans beat chance on average (Diel et al., 2024; Stockner et al., 2026). Neither models how detection changes as generators improve. The one temporal analysis available, a publication-year moderator in the face meta-analysis, found accuracy *rising* over time, the opposite of the common intuition. Its authors flag the result as exploratory and call for work mapping synthesis advances onto human performance.

A single time variable cannot separate the forces involved. Generators improve, which should lower accuracy. People gain exposure to synthetic media, which could raise it. The effort spent deploying a generator (prompting, curation, post-processing) varies from study to study. Publication year blends all three.

This paper makes three contributions:

1. A **two-clock design** that indexes each result both to the generator’s release date and to when the study was run, applied to 50 face experiments and 17 text results.
2. Pilot estimates showing a **generator-driven decay** within architecture lineages, a **reset** when a new architecture family appears, **no measurable gain from background exposure**, and, in one small study, a large gain from **hands-on expertise**.
3. A three-factor model, **Turing Decay**, with **six preregistered, falsifiable predictions** for audio and video, stated before those data were coded, and a first test of them on 125 participant groups from 58 audio and video studies.

2 Related work

Pooled human detection. Diel et al. (2024) pooled 137 effect sizes from 56 papers (86,155 participants) across audio, image, text and video, and found detection not reliably above chance in any modality. Stockner et al. (2026) focused on static faces (36 studies, 51 experiments in the meta-analysis), found 56.1% overall accuracy, lower accuracy for AI faces than for real ones, chance-level performance for StyleGAN2 and StyleGAN3, and higher accuracy in more recent publications. Our face analysis re-uses their per-experiment table.

Human detection of AI text. Clark et al. (2021) found untrained evaluators at 57.9% for GPT-2 and 49.9% for GPT-3, with brief training adding at most about 5 points. The GPT-3 report (Brown et al., 2020) measured detection of news articles across eight model sizes released together, falling from 76% to 52%. Later studies of ChatGPT-era text found teachers, linguists and lecturers near or below chance (Casal & Kessler, 2023; Fleckenstein et al., 2024; Fiedler & Döpke, 2025). Russell et al. (2025) found a sharp exception: people who frequently use ChatGPT for writing detected AI text almost perfectly. In interactive three-party Turing tests, GPT-4.5 with a humanlike persona was judged human more often than the real humans it was paired with (Jones & Bergen, 2025).

Automated detectors over time. Detectors trained on earlier generators lose accuracy on newer ones; one 2026 study reports true-positive rates of image detectors falling 16–36% between early 2023 and late 2025 (Chrysidis et al., 2026). That work concerns machines, not people, but it shows the same moving-target structure.

Gap. No prior work models human detection as a function of generator capability while separating it from observer exposure and deployment effort.

3 Pilot methods: faces and text

Face data. We took the per-experiment table (Appendix A) of Stockner et al. (2026): sample size, mean accuracy, SD and generator type. We dropped one subgroup that overlapped another experiment, leaving 50 experiments for the publication-year model. For generator-dated models we also excluded one experiment with an unidentified generator, one using face-swapping rather than synthesis, and one mixing diffusion and GAN stimuli, leaving 47. One experiment used a four-choice task; its chance level was set to 0.25.

Text data. We hand-coded 17 results from 9 studies: Clark et al. (2021), Brown et al. (2020), Gao et al. (2023), Casal & Kessler (2023), Fleckenstein et al. (2024; two studies), Fiedler & Döpke (2025), Russell et al. (2025) and Jones & Bergen (2025). Where studies reported separate hit rates for AI and human texts, overall accuracy is their mean. For three-party Turing tests, accuracy is 1 minus the rate at which the AI was chosen as human.

Coding. Each result received a generator release date (decimal year of first public release: StyleGAN 2018.95, StyleGAN2 2019.95, StyleGAN3 2021.47, Stable Diffusion 2022.64, GPT-4o image generation 2025.23; GPT-2 2019.13, GPT-3 2020.42, ChatGPT 2022.92, GPT-4 2023.20), an intervention flag (training, feedback or AI guidance), an observer-expertise flag, and the year the study was run (publication year minus 0.5 when not reported).

Model. The outcome is accuracy minus chance, in percentage points (“points” throughout). We fit random-effects meta-regressions, which weight each result by its precision and allow true effects to differ between studies (DerSimonian-Laird between-study variance with the Knapp-Hartung adjustment):

$$y_i = \beta_0 + \beta_1 \text{GenDate}_i + \beta_2 \text{YearRun}_i + \beta_3 \text{Training}_i + u_i + e_i$$

Here GenDate is the generator’s release date, YearRun the year the study was run, and Training a flag for training or feedback. β_1 is the *generator clock*: the change in accuracy per year of newer generator. β_2 is the *observer clock*: the change per year of calendar time, which captures growing public exposure to synthetic media. The cleanest test of the observer clock holds the generator fixed: we refit β_2 using only StyleGAN2 experiments, run between 2021 and 2026. Sampling variance was SD^2/n for faces; for text, where SDs are rarely reported, we used $p(1-p)/n$ and flagged these values as approximate. Code, coded data and the preregistration are archived on Zenodo (doi:10.5281/zenodo.22989632).

4 Pilot results: faces

Human accuracy on AI faces tracks the generator, not the calendar. Within the GAN lineage, each year of newer generator release cost about 5 percentage points of accuracy, while five years of studies against the same generator showed no improvement.

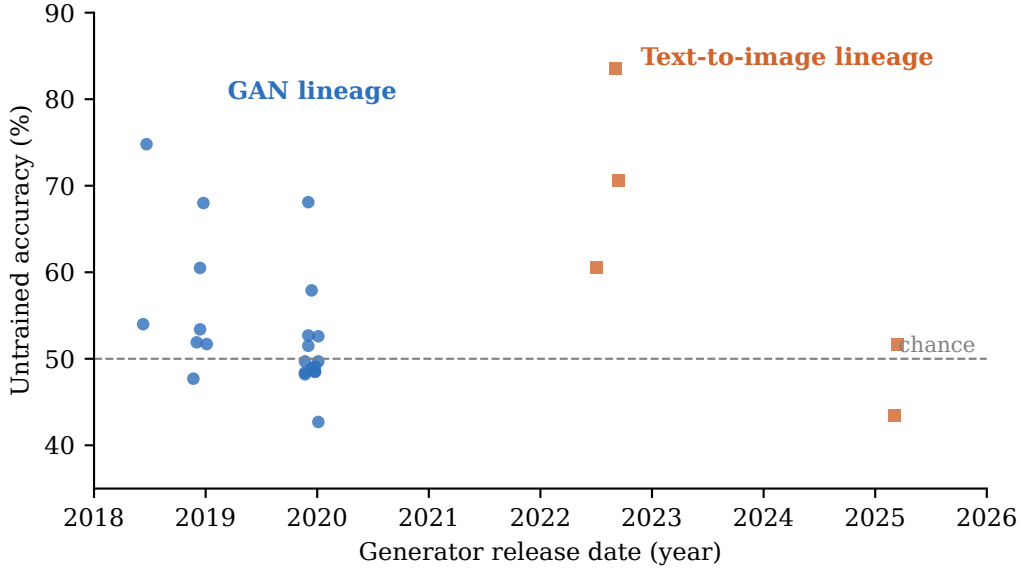


Figure 1: Each newer generator lowers accuracy; a new architecture resets it. Untrained human accuracy on AI versus real faces, one point per experiment (27 untrained experiments with a dated generator; data from Stockner et al., 2026, Appendix A). Blue circles: GAN lineage; orange squares: text-to-image lineage. Dashed line: chance.

Effect	Experiments	Estimate	p
Generator clock, GAN family	39	−5.3 pts per year of release (SE 1.8)	.006
Observer clock, same model	39	−0.8 pts per year run (SE 0.7)	.27
Observer clock, StyleGAN2 only	23	+0.2 pts per year run (SE 0.9)	.82
Generator clock, all generators	47	−5.8 pts per year of release (SE 1.4)	< .001
Text-to-image reset	47	+33 pts above the GAN trend (SE 6.4)	< .001
Training or feedback	39	+7.0 pts (SE 2.4)	.007

No observer clock was detected. Holding the generator fixed at StyleGAN2, experiments run from 2021 to 2026 show no gain in accuracy (+0.2 points per year, SE 0.9). The interval cannot rule out a gain of up to about 2 points per year. Participants in later studies had also seen other kinds of synthetic media in daily life, so holding the generator fixed does not hold their exposure fixed.

The rising trend by publication year disappears once studies are weighted properly. With publication year as the only time variable, the slope is −0.5 points per year (SE 0.8, $p = .54$). The rise reported by Stockner et al. (2026) most likely reflects later studies using diffusion-generated faces, which were the easiest to detect.

The decay is a sawtooth. Early text-to-image faces (Stable Diffusion, 2022) were detected 70–83% of the

time, far above what the GAN trend predicts for 2022. Within that lineage, accuracy then fell to 43–52% for GPT-4o-era faces (2025): about 30 points in under three years.

Training helps modestly. Training, feedback or AI guidance added about 7 points, roughly one generator-year’s worth of decay.

5 Pilot results: text

Text reached the chance floor early, and the decline follows model capability rather than time. Untrained readers caught GPT-2 text 57.9% of the time and GPT-3 text 49.9% of the time, and have stayed near chance for every generator since.

Result	Generator	Readers	Accuracy
Clark et al. 2021	GPT-2	Crowdworkers	57.9%
Clark et al. 2021; Brown et al. 2020	GPT-3	Crowdworkers	49.9%; 52%
Casal & Kessler 2023; Fleckenstein 2024	ChatGPT	Linguists, teachers	38.9%–55.4%
Fiedler & Döpke 2025	GPT-4	University lecturers	60.5%
Russell et al. 2025	GPT-4o, Claude 3.5, o1-pro	Non-experts	52.5%
Russell et al. 2025	GPT-4o, Claude 3.5, o1-pro	Frequent LLM users	94.6%
Jones & Bergen 2025	GPT-4o, no persona	Turing-test interrogators	79%
Jones & Bergen 2025	LLaMa-3.1-405B, persona	Turing-test interrogators	44%
Jones & Bergen 2025	GPT-4.5, persona	Turing-test interrogators	27%

A floor, not a slope. Across the 12 passage-judgment results from untrained readers, generator release date had no detectable slope (-0.9 pts per year, $p = .50$). This is expected once accuracy reaches chance: there is nothing left to decay. Text crossed the floor around mid-2020, about a year after GAN faces and roughly five years before text-to-image faces.

Capability, not calendar. The GPT-3 report tested eight model sizes released on the same date, which holds time constant. Accuracy fell 6.5 points per tenfold increase in parameters ($r = -0.87$, $p = .005$), and the fitted line reaches chance at about 1.8×10^{11} parameters, essentially the 175B model itself.

Expertise may help where exposure does not. Non-experts judging GPT-4o, Claude 3.5 and o1-pro text were at 52.5%. People who frequently use ChatGPT for writing were at 94.6%, and their majority vote misclassified 1 of 300 articles. This is a single study with 4 non-experts and 5 experts: one strong data point, not an established effect. It also does not show whether the advantage carries over to generators the experts have not used.

Deployment effort can push accuracy below chance. In three-party Turing tests, interrogators identified GPT-4o without a persona prompt 79% of the time, but LLaMa-3.1-405B and GPT-4.5 with a humanlike persona only 44% and 27%. GPT-4.5 was chosen as the human more often than the actual human was. Models of similar vintage spanned 52 points depending on how they were prompted.

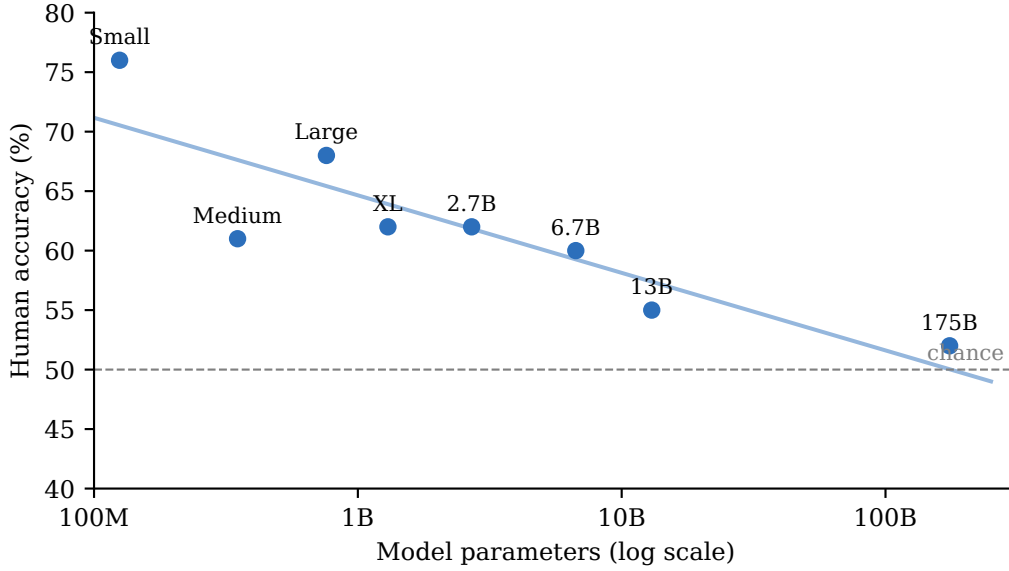


Figure 2: Same release date: detection fell 6.5 points per tenfold increase in model size. Human accuracy at spotting GPT-3 news articles across eight model sizes released together (Brown et al., 2020, Table 3.11; about 80 participants per size). Line: least-squares fit on log parameters. Dashed line: chance.

6 The Turing Decay model

We propose that the human advantage over chance, $A = \text{accuracy} - \text{chance}$, is set by three factors:

$$A = A_{0,f} e^{-\lambda_f(C - C_{0,f})} - \delta D + \gamma E$$

1. **Generator capability (C).** Within an architecture family f , the advantage decays from its starting value $A_{0,f}$ as capability rises past the family’s entry point $C_{0,f}$, at rate λ_f . Release date is a usable proxy for capability within a lineage; the size ladder shows capability itself is what matters. A new family starts a new curve, often well above the old one: a sawtooth.
2. **Deployment effort (D).** Prompting, personas, curation and post-processing lower the advantage by δ per unit of effort, and can drive it below zero, meaning people are fooled more often than chance.
3. **Observer expertise (E).** Heavy hands-on use of the generator raises the advantage by γ ; the evidence is one small study. Training helps a little. We detected no gain from background exposure through daily life.

The *half-life* of the advantage within a family is $\ln 2 / \lambda_f$: the generator progress it takes for people’s margin above chance to halve. For GAN faces an exploratory fit from the pilot suggests roughly 6 to 12 months, an order-of-magnitude estimate from few generators. For text it cannot be estimated, because the advantage had already reached zero by GPT-3. In the audio and video data (Section 9) the exploratory estimates are about 1.3 years for text-to-video and about 5 years for zero-shot voice cloning.

The model’s practical implication is that detection by eye is a transient skill tied to specific generators. If hands-on familiarity with the current generation is the only thing that keeps pace, provenance systems rather than human judgment will have to carry the weight of authenticating content.

How much of the model is tested. Of the three factors, only generator capability has solid support, from the face and text pilot and the audio d' slope. Deployment effort could not be tested, because in the published record it is confounded with the generator. Expertise rests on a single small study. The model is best read as one supported factor and two hypotheses.

7 Preregistered predictions for audio and video

The predictions below were registered in release v0.1.0 on 27 September 2026, before any audio or video accuracy value was coded; generator release dates and family assignments were then frozen in release v0.2.0, also before extraction. Each prediction applies to audio and to video separately and names the result that would refute it. They are scored in Section 10.

1. **H1, generator decay.** Within an architecture lineage, newer generators yield lower accuracy. Refuted if the release-date slope is zero or positive in both modalities.
2. **H2, sawtooth reset.** The first generator of a new architecture family is more detectable than the preceding lineage's trend predicts. Refuted if the reset term is zero or negative.
3. **H3, no background observer clock.** Holding the generator fixed, accuracy of untrained participants does not rise with the year the study was run: the 95% CI for that slope lies below +2 points per year. Refuted if the slope is reliably above +2 points per year.
4. **H4, expertise.** People with heavy hands-on use of the relevant generative tools detect better than untrained participants. Refuted if the difference is zero or negative.
5. **H5, deployment effort.** Curated, prompted or post-processed stimuli (phone-codec audio, compressed social-media video) yield lower accuracy than raw outputs of the same generator. Refuted if the difference is zero or positive.
6. **H6, modality ordering.** Video decays more slowly than audio, because it offers more channels for artifacts. Refuted if video's slope is steeper than audio's.

We also record two point predictions for later scoring: audio will still be above chance for the newest generators studied, and video will decay at under half the per-year rate observed for GAN faces.

Registered tests: random-effects three-level meta-regression (groups within papers), REML with Knapp-Hartung adjustment, one-sided $\alpha = .05$ per modality with Holm correction for H1, H2, H4, H5 and H6; H3 by the confidence-interval criterion above. A modality with fewer than 15 independent groups will be reported as descriptive only.

8 Audio and video meta-analysis: methods

We tested H1–H6 on every study we could find in which adults judged whether audio or video was real or AI-generated. The plan, including the search strings, eligibility rules and model, was frozen before the search in release v0.1.0 (doi:10.5281/zenodo.22989632); generator release dates and families were frozen in v0.2.0 (doi:10.5281/zenodo.23002314) before any accuracy value was extracted. Six deviations from that plan (D1–D6) are listed in the repository and summarised below.

Search. On 27 September 2026 we searched OpenAlex (7,814 records), PubMed (359) and IEEE Xplore (1,790) from 2016 onward, and ran backward and forward citation searches from Diel et al. (2024) and Stockner et al. (2026) (255 records). OpenAlex replaced Scopus, Web of Science and PsycInfo, which require subscriptions (D1); all searches ran through the databases' APIs so they can be rerun exactly (D3).

Screening. Two automatic rules, written before they were run, removed records with no audio, video or deepfake term (2,954) and records with no human-study term (1,043); random samples of both were read to confirm that none was eligible. The remaining records were screened by title and abstract, then by full text. An AI assistant made every screening decision and logged a one-line reason for each; the author did not review each decision (D2). Records the assistant could not decide from the abstract went to full text rather than being excluded.

Eligibility. Adults judged real versus AI-generated audio (speech, voice, music) or video; the task contained both real and generated stimuli; and the report gave accuracy, hit and false-alarm rates, d' , or values from which they could be computed. We excluded quality or credibility ratings without a real-versus-fake judgment, non-generative manipulations, and non-empirical reports.

Extraction. One row per independent participant group, split by generator when a study reported accuracy per generator. Every value carries a verbatim quote and its location. When a study reported only hit and false-alarm rates, we used balanced accuracy, $(\text{hit rate} + \text{correct-rejection rate}) / 2$. Extraction was also done by AI assistants, reading full texts partly through a summarising web tool (D4); 38 of the 40 studies in the model were then re-checked against their sources, and three studies were dropped at that stage (a majority-vote metric, a duplicate publication, and point-light stimuli). Authors were not contacted (D6). Instead, missing values were recovered from full texts and preprints read in a web browser or downloaded from open-access pages, from the authors' posted trial-level data (three studies), and, where only a figure reported them, from the printed figure values. Each recovered value is flagged with its method.

Generator coding. Each group was mapped to the frozen generator table using only the generator names. Generators missing from the table (51) were added under the frozen rules and dated from arXiv v1 dates, repository histories or product announcements; 33 of these dates were confirmed against a primary record and the rest are marked medium or low confidence. Pipelines such as Tacotron 2 + HiFi-GAN were dated by their newest component, and audiovisual groups were analysed with video (D5).

Model. Accuracy minus chance was modelled in a three-level random-effects meta-regression (groups within papers), fitted by REML with t-tests on $k - p$ degrees of freedom, separately for audio and video. The full model includes generator release date, family reset, deployment effort, observer expertise, year run and training; H6 adds a release-date $\times$ modality interaction to the pooled model.

Of the 7,345 records screened out, 3,997 were removed by the two automatic rules and 3,348 by title and abstract. Full-text exclusions were 31 with no real-versus-fake judgment (E4), 25 not audio or video (E2), 11 not empirical (E3), 7 with no human judgment (E1) and 1 non-generative (E5). Of the 90 included studies outside the model, 54 report no usable numbers, 13 name no datable generator, 10 report a metric other than accuracy, 6 are music, non-speech audio or non-face video (exploratory only), 1 cannot separate minors, and 6 were removed at extraction or verification (3 ineligible on full reading, 3 dropped).

9 Audio and video: results

People still beat chance on both audio and video (chance is 50% in almost every study): by about 17–18 points for generators released in 2020, and by about 12 points for the newest audio generator (2024) and 16 for the newest video generator (2025). For audio, detection declines with newer generators when measured as sensitivity (d' , a secondary measure), and only weakly when measured as accuracy; video shows no overall decline. These results are consistent with the model but not confirmatory. They come from 125 participant groups in 58 studies (58 audio groups from 27 studies; 67 video or audiovisual groups from 32). In the table, a negative slope means people do worse against newer generators.

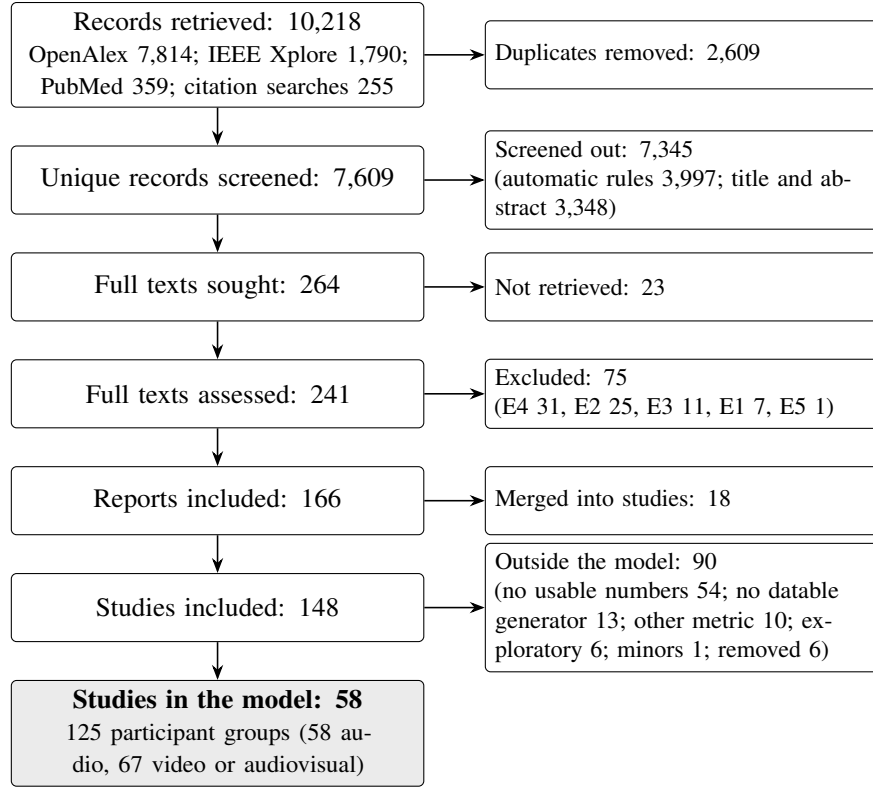


Figure 3: PRISMA 2020 flow for the audio and video search of 27 September 2026.

Test (accuracy points per year, 95% CI)	Audio	Video
Accuracy above chance, generator released in 2020	+17.7	+16.8
H1 generator release date, alone	−1.3 [−3.7, +1.2]	−0.1 [−1.2, +1.0]
H1, untrained public only	−0.1 [−3.0, +2.8]	+0.1 [−1.1, +1.3]
H1, full model	−1.6 [−5.9, +2.8]	+0.1 [−1.3, +1.6]
H1 in d' per year	−0.23 [−0.43, −0.04]	−0.01 [−0.10, +0.08]
H3 year the study was run, generator held fixed	−0.3 [−9.5, +8.8]	+1.8 [−0.8, +4.4]

H1, generator date. For audio, accuracy falls about 1.3 points per year of newer generator (one-sided $p = .15$), and sensitivity falls 0.23 d' per year, an interval that excludes zero. d' is a secondary measure, not the registered primary test, which used accuracy, and video d' is flat. For video the slope is flat. Pooling both modalities, the slope is $-1.2 [-3.2, +0.8]$ points per year (one-sided $p = .11$). None of the directional tests in the full model survives the registered Holm correction for multiple tests (audio H1 adjusted $p = .71$).

The audio accuracy slope was steeper (-2.1) before the last six studies were added. Two of those studies had listeners judge clones of their own or a friend's voice made with recent generators, and they scored 86–90%. Dropping every familiar-voice group still leaves the slope at $-1.6 [-4.3, +1.2]$, so familiarity does not explain the change.

H3, observer clock. Holding the generator fixed, accuracy shows no reliable trend with the year a study was run. Neither interval yet excludes the registered +2 points per year, so H3 is not supported by the equivalence criterion.

H6, modality. Video declines more slowly than audio, as predicted: the video-minus-audio slope is +1.1 [−1.3, +3.5] points per year, in the predicted direction but not significant.

H2, H4, H5. Family resets, deployment effort and expertise remain hard to test: only five groups were experts, and few studies processed their stimuli (for example, through a phone codec) or said how they selected them. Estimates are in Section 10.

Decay within a lineage. Fitting the above-chance margin as an exponential decay within each lineage (registered as a secondary estimate; exploratory) shows the sawtooth most clearly. Within text-to-video, people start about 33 points above chance and the margin halves in about 1.3 years (18 groups, 14 release dates, 2023–2026). Within zero-shot voice cloning it starts at 26 points and halves in about 5 years (30 groups). Face-swap autoencoders and early neural TTS show no decay across their few dates.

Who listens matters. In one study, d/Deaf viewers caught 41% of audio-only fakes against 90% for hearing viewers, while all groups reached 84–87% on fully audiovisual fakes (Pasternak et al., 2026). Whether a fake is detectable depends on the channels an observer can use, not only on the generator. These are observer characteristics (sensory channel, familiarity with the voice), not hands-on generator use, so they do not support the expertise term E as defined; they point to a broader observer term for a future version of the model.

9.1 Sensitivity analyses

The audio accuracy slope is negative under every check except exact variances only (19 groups), but it is not robust to dropping influential groups, which leaves −0.4 points per year; the d' slope is the more stable result. Twelve groups used single-item tasks with more than 25% more stimuli of one class; using balanced accuracy for seven of them and dropping five leaves both slopes essentially unchanged. The video slope stays at zero throughout.

H1 slope under (points per year, 95% CI)	Audio	Video
Primary, release date alone	−1.3 [−3.7, +1.2], k = 58	−0.1 [−1.2, +1.0], k = 67
Influential groups dropped	−0.4 [−3.3, +2.5], k = 54	+0.2 [−1.0, +1.4], k = 64
Leave one paper out (range)	−2.3 to −0.4	−0.4 to 0.0
Exact variances only	+2.2 [−4.0, +8.5], k = 19	−1.0 [−3.4, +1.3], k = 13
Approximate sample sizes excluded	−1.7 [−4.3, +0.8], k = 54	−0.4 [−1.5, +0.8], k = 65
Dataset-coded groups excluded	−1.4 [−4.5, +1.7], k = 47	−0.4 [−2.0, +1.2], k = 42
Familiar-voice groups excluded	−1.6 [−4.3, +1.2], k = 43	+0.1 [−1.1, +1.4], k = 48
Unequal real/fake counts: balanced accuracy or dropped	−1.3 [−3.8, +1.3], k = 55	−0.3 [−1.4, +0.9], k = 65
Audiovisual groups removed	—	−0.3 [−1.3, +0.7], k = 48
d' instead of accuracy (d' per year)	−0.23 [−0.43, −0.04], k = 31	−0.01 [−0.10, +0.08], k = 41

Replacing the year a study was run with its publication year leaves H3 unchanged (audio −2.2 [−7.4, +3.0];

video +0.9 [−1.4, +3.1]). Egger’s test shows no clear small-study effect in either modality (audio $p = .14$; video $p = .89$). The registered step-function selection model was not run because there are too few groups per modality.

10 Scoring the preregistered predictions

The table scores each prediction against the refutation criterion stated in Section 7, using the full registered model for H1–H5 and the pooled model for H6. Counts in brackets are the number of groups that carry each contrast.

Prediction	Audio	Video	Verdict
H1, generator decay	−1.3 [−3.7, +1.2] pts/yr; −0.23 [−0.43, −0.04] d’/yr	−0.1 [−1.2, +1.0] pts/yr	Not refuted: audio slope negative, significant in d’ only; not significant after Holm
H2, sawtooth reset	−7.1 [−27.3, +13.1] (7 groups)	+2.9 [−5.0, +10.8] (12 groups)	Not supported; too few reset groups
H3, no observer clock	−0.3 [−9.5, +8.8]	+1.8 [−0.8, +4.4]	Inconclusive: neither upper bound is below +2 pts/yr
H4, expertise	+20.4 [−7.3, +48.0] (3 groups)	−8.8 [−23.4, +5.7] (2 groups)	Not testable: too few expert groups
H5, deployment effort	+3.2 [−4.6, +11.0] (12 groups)	−2.5 [−7.5, +2.5] (36 groups)	Not supported; video in the predicted direction (one-sided $p = .16$)
H6, video decays more slowly	—	video minus audio +1.1 [−1.3, +3.5] pts/yr	Direction as predicted, not significant
Point: audio above chance at the newest generator	+11.7 [+3.9, +19.4] pts at 2024.7	—	Supported
Point: video decays at under half the GAN-face rate (2.7 pts/yr)	—	−0.1 [−1.2, +1.0] pts/yr	Supported

11 Discussion

The claim these data support is narrow: within a generator lineage, human detection falls as generators improve and does not recover with calendar time, and hands-on expertise may counteract it. The face and text pilot carries most of that claim. The preregistered audio and video test is consistent with it but underpowered: the audio decline is clear only in sensitivity (d’), the accuracy slope shrinks when influential groups are dropped, and nothing survives the correction for multiple tests. We see that as the paper’s main value: a preregistered pilot that shows where the evidence is strong and where it is not. For audio, d’ is also the better measure of the generator clock, because accuracy mixes discrimination with response bias, and warnings,

nudges and “half the clips are fake” instructions shift how often people answer “AI” without changing how well they tell the classes apart (Kirk, 2026).

Why faces and text differ. Untrained readers reached chance on text by GPT-3 in mid-2020, about a year after GAN faces reached it with StyleGAN2; text-to-image faces then reset the curve and fell again. Two differences may explain why faces keep a measurable slope while text sits on a floor. Judging a face is a perceptual task with physical cues such as skin texture, lighting and symmetry, which each generator removes a little more, while judging a passage relies on style and plausibility, which language models matched early. Face studies also keep reusing older generators, so the face record samples more of each lineage’s decline. Both explanations are testable, and neither is established here.

Video shows no decline when all lineages are pooled, yet the fastest decay in the whole dataset sits inside video: text-to-video generators began about 33 points above chance in late 2023 and lost half that margin in about 1.3 years. Pooling hides this because much of the video literature still tests face-swap datasets from 2018–2020, which people detect at a steady, modest rate. This is the sawtooth that Turing Decay predicts: each new family starts detectable and decays, and a single pooled slope averages over several teeth.

What an observer brings matters as much as the generator. Listeners who heard clones of their own or a friend’s voice caught them 86–90% of the time, against chance for strangers hearing the same clones (Rosi et al., 2025; Sanchez et al., 2026), and d/Deaf viewers caught 41% of audio-only fakes against 90% for hearing viewers (Pasternak et al., 2026). Detection is a property of the observer, the channel and the generator together, not of the generator alone. This is the paper’s strongest observer-side evidence, and it concerns channel and familiarity, not the hands-on use that E measures.

The model’s other factors could not yet be tested. Only five groups were experts, few studies report how stimuli were selected or processed, and family resets are identifiable only where a study tested the first release of a new architecture. These gaps are features of the literature rather than of the model. Future studies would make the question answerable by reporting the exact generator and version, the dates of data collection, hit and false-alarm rates separately, how stimuli were selected and processed, and by including participants with hands-on generator experience.

The practical conclusion is unchanged from the pilot. Human detection is still above chance on average, but the margin is fragile: it is smallest for the newest generators, can be erased by careful selection and prompting, and is largest for people who know the voice or the tools. Detection by eye and ear is a transient skill tied to specific generators, and it cannot be the main defence against synthetic media.

12 Limitations

The face and text pilot rests on thin data in several places:

- **Few generators.** The face slope rests on about six distinct generators, and StyleGAN3 appears only in trained groups. The text-to-image lineage has two untrained experiments per generator.
- **Hand coding by one author.** Release dates, intervention flags and expertise codes were assigned by the author alone, without a second coder. Study quality was not included as a moderator for faces.
- **Approximate variances for text.** Most text studies report only percentages, so sampling variances use $p(1-p)/n$, which ignores trials per participant and within-person correlation. The Turing-test sample sizes per witness are approximate.
- **Small expertise comparison.** The expert-versus-novice contrast rests on 9 annotators in a single study.
- **Coarse generator labels.** Several text studies identify the generator only as “ChatGPT”.
- **Stimulus curation.** Many studies select the most realistic outputs, and few report how. Deployment

effort is therefore confounded with generator in the published record, which is why the audio and video analysis coded it explicitly (H5).

- **Publication year as a proxy.** Where the data-collection year was not reported, we assumed studies ran six months before publication.

The audio and video meta-analysis has limitations of its own:

- **Missing studies.** 54 of the 148 included studies report no usable numbers and could not be analysed without contacting authors (D6). If studies with poor or null results report less, the estimates are biased in unknown directions.
- **Recovered and derived values.** Some values were read from figures, computed from posted trial data, taken from a model-based estimate (one study), or entered with approximate sample sizes (one study counts responses, not people). Each is flagged, and sensitivity analyses excluding approximate values move the audio slope by less than half a point per year.
- **AI-assisted extraction and coding.** Extraction and generator coding were done with AI assistants and re-checked against sources by one author, without an independent second coder. In a paper about detecting AI-generated content this is a real concern; a second human coder, starting with the expertise and familiarity codes, is the most important next step.
- **The two clocks are not fully separable.** Holding the generator fixed does not hold exposure fixed: participants in later studies had seen other kinds of synthetic media in daily life. The observer-clock test therefore measures exposure to the media environment as a whole, not to one generator.
- **Reset or quality?** Early text-to-image faces may have been easier to spot partly because their images were rougher, not only because they came from a new family. The reset term cannot separate the two.
- **Expertise transfer.** The one expertise contrast does not show whether heavy users of one generator detect the output of others as well, or only the artifacts they already know.
- **Base rates.** Twelve groups used single-item tasks with more than 25% more stimuli of one class, where accuracy minus 50% overstates or understates the margin. Using balanced accuracy for seven and dropping five leaves both slopes essentially unchanged (Section 9).
- **Generator dates.** 51 generators were added to the frozen table; 18 of their dates are medium or low confidence. Dataset-coded groups pool several generators under one date.
- **Thin lineages.** Half-lives rest on few release dates per lineage, and heterogeneity between papers is large relative to the slopes being estimated.
- **Deviations.** Six deviations from the registered plan (D1–D6) are listed in the repository, most importantly that authors were not contacted.

The analysis will be rerun as the missing studies’ values become available.

13 Data and code availability

The preregistration (release v0.1.0, doi:10.5281/zenodo.22989632), the frozen generator table (release v0.2.0, doi:10.5281/zenodo.23002314), and the audio and video analysis with the manuscript (release v0.3.0, doi:10.5281/zenodo.23010813; revised in v0.3.1, doi:10.5281/zenodo.23049109, and v0.3.2) are archived on Zenodo; the concept DOI 10.5281/zenodo.22989631 always resolves to the latest version. The extraction sheet with verbatim quotes, the search and screening records, the deviation log and all analysis code are at github.com/jbussin/turing-decay.

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